



OPEN DengueGNN: Graph-based deep learning for modeling disease spread dynamics and prediction

Rasitha Banu GulMohamed¹✉, Wafa Hetany², Hanan Abdullah Almaimani^{3,4} & Faiza Abdalla Saeed Khiery⁵

Dengue fever is a mosquito-borne disease that is rapidly spreading across the world due to climate, human mobility and other regional connections, making it a serious challenge for public health. Statistical and machine learning models that are commonly used do not adequately represent the complex spatio-temporal patterns of disease transmission and thus produce poor forecast and delayed response for dengue transmission. This study proposes Dynamic Spatio-Temporal Graph Neural Network (ST-GNN), which accurately predicts dengue fever regionally. The model produces a dynamic graph whose nodes are regions, the human mobility or adjacency is represented by edges, and other environmental and historical epidemiological characteristics are used for the graph nodes. Graph convolution layers are used for representing spatial dependence and attention augmented LSTMs are used for representing temporally evolving data. The graph convolution and attention layers make it possible to fuse the environmental and mobility features for the purpose of enhancing predictions. Experimental validation done on OpenDengue dataset indicates that ST-GNN outperforms the 10 baseline models with RMSE of 6.1, MAE of 4.9 and MAPE of 12.0% for a one-week prediction and RMSE of 10.9, MAE of 8.8 and MAPE of 22.4% for a four-week prediction with minimal RMSE observed for one-week prediction and highest correlation in space (Moran's I: 0.76 and 0.68 respectively). To validate the importance of dynamic graphs, temporal attention and mobility features, we performed ablation experiments. In addition, the GNNExplainer and Integrated Gradients showed a region with high risk and key environmental drivers. Taken together, the proposed framework enables the graspable multi-horizon forecasting to guide proactive dengue prevention and corresponding public health interventions.

Keywords Dengue prediction, Spatio-temporal graph neural networks, Human mobility, Environmental factors, Dynamic graph modeling, Explainable AI, Multi-horizon forecasting

Dengue is a disease transmitted by the Aedes mosquito, and it remains one of the most important public health problems in tropical and subtropical areas, infecting millions of people annually¹. Dengue occurrence was identified as a complex process that is influenced by environmental factors (e.g., temperature, rainfall, and humidity) and human activities (e.g., human mobility and urbanization)². Therefore, proper prediction of dengue incidence is required for appropriate public health response, preparation for health care utilization, and reduction of outbreak-related socio-economic burden³. Unfortunately, contemporary statistical approaches (e.g., ARIMA, SARIMA)⁴ are unsuited for modelling nonlinear temporal processes and interdependence between regions while traditional machine learning methods (i.e., Random Forest, XGBoost) are unsuitable in modelling spatiotemporal spreading processes involving vector-borne disease⁵.

With the advance of deep learning technology, such as recurrent neural network (RNN)⁶, long short term memory (LSTM)⁷ and gated recurrent unit (GRU)⁸, the modeling ability of temporal dynamics for epidemiological data is enhanced. The convolutional LSTMs and further spatio-temporal diffusive network, integrate information of local neighbourhoods and deep layers and connections to key nodes of spatio-temporal objects into the network⁹. However, we can see these models as fixed spatial relationships that limit adaptation

¹Department of Public Health, Health Informatics Program, College of Nursing & Health Sciences, Jazan University, Jazan, Saudi Arabia. ²Department of Public Health, College of Nursing and Health Sciences, Jazan University, Jazan, Saudi Arabia. ³Department of Public Health Epidemiology, College of Nursing and Health Science, Jazan University, Jazan, Saudi Arabia. ⁴Master in Public Health Epidemiology, Toledo University, Ohio, USA. ⁵Department of Public Health, Health Education and Promotion Program, College of Nursing and Health sciences, Jazan University, Jazan, Saudi Arabia. ✉email: rbanu@jazanu.edu.sa; rashidabanu0476@gmail.com

to naturally occurring dynamic human movement and the ever-changing human-environmental factors that are important for dengue transmission. Instead, as shown in Fig. 1 presents the Graph Neural Networks (GNNs) can represent regions as interconnected nodes, with relations being other nodes.

This study aims to address these problems by formulating a Dynamic Spatio-Temporal Graph Neural Network (ST-GNN) that explicitly captures the time-varying inter-regional connectivity considering environmental and mobility conditions¹⁰. The dynamic graph representation enables for an adaptive or responsive update for node connectivity representing real-world evolution for movement and adjacency of human beings over time. Spatial dependencies are learned using graph convolution layers while time evolution is represented using LSTM augmented with attention which characterizes the behavior in history which is the most relevant for prediction. Environmental and mobility features are combined with spatio-temporal embeddings to form a network that can learn complex associations between climate, movement and the trajectory history of human in the pursuit of the disease¹¹. For explanation, we use GNNExplainer and Integrated Gradients, thus giving actionable interpretive insight for supporting public health decision making¹².

The main research question answered in this study is as follows: Can a data-informed dynamic spatio-temporal graph neural network that integrates environmental and mobility data produce accurate and interpretable multi-horizon dengue incidence predictions at a regional scale? In addition, the research question is a step in the direction from generic problem of dengue forecasting to focusing on a particular shortcoming of the current dengue forecasting model for joint representation of dynamic spatial connectivity, temporal change, and auxiliary drivers. The study objectives are as follows:

- To develop a dynamic graph-based spatio-temporal model that will capture the time varying inter-regional connectivity for predicting dengue
- To employ environmental data and human mobility data with spatio-temporal embeddings for better forecast accuracy.
- To create an interpretable multi-horizon forecast for public health interventions and policy decision making

The major contribution of the study is the first large-scale, dynamic, spatio-temporal modeling of transmission of dengue in a multi-country surveillance dataset (OpenDengue). The novelty of the proposed ST-GNN is therefore related to an application-driven methodology design, that is composed of: (i) dynamic graph construction considering the presence of geographic adjacency and gravity-based human mobility, (ii) temporal attention learning under epidemic regime changing, (iii) uncertainty-aware learning, and (iv) methodic validation under heterogeneous epidemiology scenarios.

The remainder of this paper is organized as follows: Section "Related works" is a review of the related works in the field. Section "Method" introduces the proposed ST-GNN model and the components of ST-GNN, including graph construction, spatial and temporal encoder, and feature fusion. Section "Experiments" describes the dataset, experimental setup, evaluation measures, and baseline comparisons. Section "Experiments" discusses the result, ablation studies and explainability analyses before concluding remarks and future research directions presented in Section "Conclusion and future recommendations".



Fig. 1. Graph neural network for disease prediction.

Related works

There has been great interest in modeling and forecasting disease dynamics in many health contexts using graph-based methods, such as Graph Neural Networks (GNNs) and other graph theoretic approaches. These approaches build upon their ability to capture the underlying relational structure of clinical, epidemiological, and mobility data to encode complex spatial, temporal, and inter-personal correlations that traditional approaches are not well suited to do so. Recent work has established the effectiveness of graph-based approaches for tasks, such as predicting dengue case severity and case forecasting.

Dhote et al.¹³ developed a GNN with Attention Mechanism (GNN-AM) for predicting dengue case severity, including patient's clinical indicators for greater accuracy. This study demonstrated better prediction performance compared to statistical models. Importantly, real-world implementation, while promising, might be hindered by challenges involving integrating multiple data sources, generalizability to other settings, scalability in other settings, and privacy-preserving¹³.

In another study analyzing dengue, Weng et al.¹⁴ also demonstrated GNNs were an innovative method for dengue forecasting in Sri Lanka by integrating different forms of spatiotemporal, virology, and vector data on vector and dengue cases. They demonstrated an early warning system based on a GNN model outperformed various traditional methods (e.g., ARIMA, Random Forest, LSTM). It is important to note though, while the GNN model worked well, it still may struggle with data quality, time period coverage, or scalarability in other contexts¹⁴.

Wang et al.¹⁵ present a Metapopulation Graph Transformer Neural Network (M-Graphormer) combining GNNs with SIR models for multi-regional infectious disease prediction. The framework captures hidden spatial dependencies and human mobility effects, accurately estimating high-dimensional parameters. Critically, while robust, its performance may vary with real-time data quality and the complexity of intervention dynamics¹⁵.

Liu et al.²⁸ assess the use of Graph Neural Networks (GNNs) in epidemic forecasting, providing differentiations in task and approach, including neural and hybrid methodologies. They also point out the potential advantages of GNNs over traditional mechanistic models in accounting for complex transmission processes. However, they note important challenges related to data sparsity, model explainability, and functioning within the wider context of public health systems¹⁶.

Fritz et al.¹⁸ leverage graph neural networks with spatio-temporal models of disease spread to here predict weekly cases of COVID-19 in Germany, including human mobility data and relation. They present a multimodal forecasting framework which allows more accurate and interpretable predictions. However, while the framework may be seen as effective, its robustness and applicability could be affected by the level of data available, as well as complexity of the models, and to what extent applicability is limited to other regions or across outbreaks¹⁷.

Alrashdi and Taloba¹⁷ implement a hybrid model of GNN with LSTMs to predict heart failure diagnosis, fully reflecting the complex interdependencies of clinical variables alongside time. The GNN-LSTM model was evaluated and shown to achieve 98.9% accuracy on a representative sample of a large clinical dataset and outperforming a large cohort of traditional models. However, while highly effective, further testing is necessary to determine generalizability across a broader population and healthcare contexts¹⁸.

Zhu et al.¹⁹ proposed the Graph Attention-based Spatial Temporal (GAST) model for epidemic forecasting, incorporating spatial and temporal dynamics through graph attention networks. The study showed that the model outperformed commonly used traditional approaches for forecasting short-term influenza and COVID-19 observations. Importantly, challenges remain surrounding model interpretability, issues related to data quality, model transferability and overfitting, and implementation occurs out of an experimental context. Hence, further testing is called for to ensure a wider scope of application¹⁹.

In a review of existing works in the field, Lu et al.²⁰ discussed graph machine learning (ML) approaches to disease prediction while utilizing electronic health data and thus noted approaches focusing foremost, on node classification, and link prediction. The work highlighted the unique advantage of GNN approaches in that they generally outperformed traditional ML approaches without graphs. However, aspects of the works still present various challenges pertaining to: clinical applicability and relevance, intuitive interpretability, and exploiting dynamic graphs rich with relationships from the health data²⁰.

Furthermore, Adamopoulos et al.²¹ applied applications of graph theory algorithms to model COVID-19 transmission, utilizing measures of centrality, community detection and epidemic spreading models, to establish useful pathways of transmission or possible intervention. Limitations were however present, in recognizing varied potential trajectories of transmission by individuals and utilization of graphs and estimates were hampered by data quality, scalability and efforts to generalize the theoretical discoveries into public health strategies or efforts²¹.

Even though graph-derived models have exhibited superior predictive performance in disease forecasting and clinical prediction, each still has considerable limitations. Many of the studies that underlie graph-based models depend heavily on well-developed, high-quality, and comprehensive datasets, thus making performance suitable only if data is not missing or noisy. Generalizability across regions or populations, or within different disease contexts, is still unknown and unclear, as many graph models have been designed and trained with specific geographic or clinical domains in mind. Further, while the interpretability of models is a challenge in most cases, it is especially challenging for complex hybrid architectures, such as those that combine GNNs with temporal models or mechanistic models. In addition, scalability and real-time deployment within an active and operational healthcare system is largely untapped. Furthermore, dynamic or evolving graphs, or other less-defined graphs, have a unique set of issues that require consideration. These limitations are essential to address in order to enable translation of these models to meaningful public health and clinical applications.

Method

This study uses a graph-based deep learning methodology with spatial, temporal, environmental, and mobility features for prediction of dengue outbreaks. The methodology defines areas as dynamic graphs that capture spatial dependencies through graph convolutions, while extracting temporal dynamics using attention-enhanced LSTMs. An integrated fusion module combines heterogeneous features and the prediction layer provides forecasts for each area. Techniques for optimizing the methodology are specified and explainability tools are provided to ensure the robustness, interpretability, and epidemiological relevance of the methodology. Reliability, actionable implications of the predictions and transparency make the methodology adaptable for robust, transparent prediction of dengue for public health.

Graph construction

Graph construction represents the study region as a dynamic graph $G_t = (V, E, X_t)$ that captures both spatial and temporal dependencies for predicting dengue spread. Here, nodes V represent regions or districts, and edges E represent adjacency or mobility, such as human travel, roads, or human mobility, representing the pathways for dengue to spread. The matrix of features $X_t \in R^{N \times F}$ stores attributes at time t , including environmental attributes (temperature, rainfall), density of population, and prior cases of Dengue for each node. This dynamic graph facilitates the model to leverage change over time as environmental conditions and mobility patterns change. The parameter X_t is given by Eq. (1)

$$X_t = [x_t^1, x_t^2, \dots, x_t^{n_1}]^T \quad (1)$$

where $x_t^i \in R^F$. This framework establishes the basis for spatio-temporal modeling, which will allow succeeding graph neural networks to learn dependencies across both nodes and time, so that the spatial propagation and temporal evolution of dengue outbreaks is simultaneously captured²². Figure 2 depicts the Framework for spatio-temporal dengue prediction using graph neural networks.

Specifically:

- Edges are defined by a hybrid formulation, i.e. in combination of geographic adjacency and human mobility flow.
- A gravity model based mobility prior is used in estimating the inter-regional transmission strength.
- Edge weights are normalized using symmetric normalization (i.e. row normalized).
- The graph is then sparsified using the concept of a top-k threshold per node in order to keep the epidemiologically meaningful connections.
- The adjacency matrix is changed on a weekly basis to account for dynamic changes in mobility and environment.

The study region is modeled as a time varying weighted graph $G = (V, E, X)$, where nodes V represent districts and edges E_t encodes dynamics transmission potential derived from both geographic proximity and human mobility. The dynamic adjacency matrix, $A_t \in R^{N \times N}$ is defined by the Eq. (2).

$$A_t = aA^{geo} + (1 - a)A_t^{mob}, \quad (2)$$

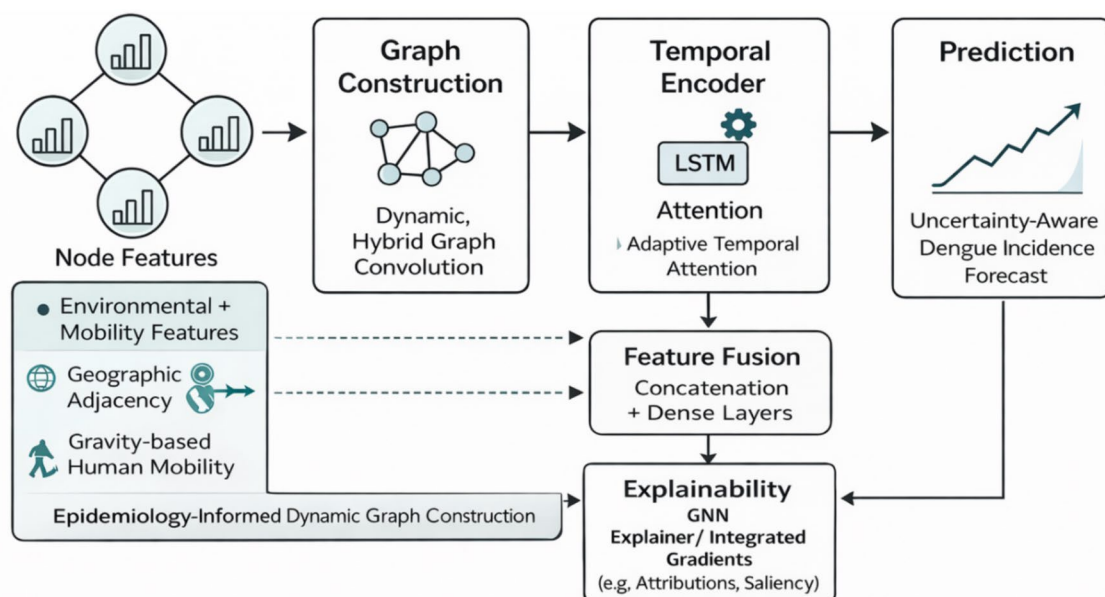


Fig. 2. Graph based Modeling for Dengue Spread Modeling.

Where A^{geo} is the static geographic adjacency matrix (binary or distance – decayed) and A_t^{mob} is a time-dependent mobility matrix, and a $\epsilon[0,1]$ controls the spatial -mobility tradeoff.

The mobility matrix A_t^{mob} is computed using a gravity model defined by the Eq. (3)

$$G_{ij}(t) = \frac{p_i p_j}{d_{ij}^2} \quad (3)$$

where p_i, p_j are the population of regions i and j , and d_{ij} is the geographic distance. When real mobility data is available, $G_{ij}(t)$ is multiplied by observed mobility flow $M_{ij}(t)$

The adjacency matrix is symmetrically normalized and it is represented by the Eq. (4)

$$A_t = D_t^{-\frac{1}{2}} A_t D_t^{-\frac{1}{2}} \quad (4)$$

where D_t is the degree matrix of A_t .

Spatial encoder (graph convolution layer)

The spatial encoder identifies spatial relationships among nodes based on adjacency matrix A . It points out the spatial relationships across regions through sharing of information through a graph convolution process across nodes. Let the adjacency matrix $A \in R^{N \times N}$ and node features $X_t \in R^{N \times F}$, the graph convolution layer computes using the Eq. (5)

$$H_t = \sigma(A \cdot X_t \cdot W_s) \quad (5)$$

where W_s is a trainable spatial weight matrix and σ is an activation function (ReLU). In this operation features from adjacent nodes go through an aggregation process, where the adjacency structure provides a weighting scheme for the aggregating process. The joint node representation mathematically takes into account the concept of local spatial context. For incident dengue prediction, this helps to ensure that areas connected to, or traveling by people, are sufficiently included in the learning process. By obtaining the spatial temporal patterns that the model is able to distinguish how a neighbouring outbreak could be partially dependent, or at least influence beginning of outbreaks in adjacent areas of space that have not previously reported an outbreak. The down-edges in the graph could be stacked together to take multi-hop move through the graph and the multi-hop frameworks are introduced to consider the local and neighbourhood distance dependencies. Thus, spatial dependencies can be modeled by including the relations between spatial areas that might be identified at some distance, which is relevant for efficient prediction of spatially correlated disease processes²³.

Edges in the dynamic graph are represented by a hybrid formulation, that is, a combination of geographical adjacency and the human mobility flows. A prior of gravity-based mobility is used to estimate the inter-regional transmission strength where population sizes, travel distance and observed mobility indices determine the flow intensity between regions. This formulation allows the model to capture the actual disease diffusion pathways in the real world that are driven by human movement and not only static connectivity between countries.

Temporal encoder (Temporal attention/LSTM)

The temporal encoder captures the temporal dynamics in dengue incidence. It describes dengue changes over time through sequential historical data. For each node, it uses spatial representations from the previous L time steps, $H_{t-L:t}^i$ are fed into an attention enhanced LSTM and it is represented by the Eq. (6).

$$h_t^i = LSTM_{attn}(H_{t-L:t}^i) \quad (6)$$

The LSTM model temporal dependencies, while the attention mechanism gives positive weights to the time steps with significant temporal trends relevant for predicting events of interest. As the LSTM architecture enables the model to pay attention to significant past events (e.g., recent increases in dengue and seasonality), the temporal encoder will retain the short-term variance, while long-term trends will be reflected in the predictions that in turn will yield a robust model for dengue temporal variability. The combination of the sequential models and the attention allows the model to dynamically pay attention to the most important, past features, for the history of dengue in the particular region. This is representing significant improvement in the focus of the model accuracy while also providing valuable, meaningful multi-step estimates. Let the history of log of each node i , which is defined as the set of characteristics of the node from the past L -length window with the Eq. (7)

$$H_{t-L:t}^i = [H_{t-L}^i, H_{t-L+1}^i, \dots, H_t^i] \quad (7)$$

Then the sequence in Eq. (7) is passed through an attention-enhanced LSTM and it is given by the Eq. (8).

$$H_t^i = LSTM_{attn}(H_{t-L:t}^i) \quad (8)$$

The attention mechanism assigns weights to previous time steps based on which time step(s) are most relevant to the prediction of dengue incidence. Even though in terms of predicting future patterns, nodes that have similar patterns in the past start from the same point, attention weights this influence in a dynamic manner²⁴.

Temporal dependencies are modelled using the attention enhanced LSTM. For each of the regions, a sliding window of historical representations is fed to the LSTM, and an attention mechanism is used to assign innovative weights to each time step. This helps the model to concentrate on important moments such as outbreak onsets,

seasonal peaks and recent surges so that modeling the time variability of dengue incidence can be done more accurately.

Integrated mobility & optimization module

This component combines environmental and mobility functionalities with spatio-temporal embeddings for better prediction capability. This combination makes it possible for this model to simultaneously account for local temporal dynamics, environmental factors, and human mobility patterns fueling the breakthroughs of dengue Fever. By incorporating heterogeneous features the network can learn complex interactions, whereby increased rainfall, and increased mobility, jointly increases transmission risk. The ReLU activated dense layer optimizes the fused representation by creating rich representations of the spatial, temporal, and auxiliary information and it is an important contribution to create a solid foundation for future region specific dengue estimation given by the Eq. (9)

$$Z_t^i = \text{ReLU}(|h_t^i|e_t^i| m_t^i] . W_f) \quad (9)$$

where h_t^i is the temporal embedding, e_t^i environmental features(e.g., temperature, rainfall), m_t^i mobility features(e.g. human movement), $[|]|$ denotes concatenation and W_f is a trainable weight matrix. This combination allows the model to simultaneously account for local temporal dynamics, environmental factors, and human mobility patterns which are all important drivers of dengue outbreaks^{25,26}.

Prediction layer

The third layer gives the model the power to translate what it has learned about spatio-temporal patterns and patterns in the environment into what can be used for a prediction. By creating predictions by region, it allows for public health to occur with precision. This layer is flexible and can be extended to probabilistic output or confidence intervals. It is simple and thus easily interpretable, while it is related to rich fused embeddings to preserve prediction accuracy for land dynamic dengue outbreaks. The fusion is then used for generating fused embeddings from which the prediction layer generates dengue incidence predictions and it is given by the Eq. (10)

$$\hat{y}_t^i = Z_t^i . W_p + b_p \quad (10)$$

where W_p and b_p are trainable parameters. Each node's fused embedding Z_t^i is mapped to a scalar output representing predicted cases. For multi-horizon forecasting, separate output layers can produce predictions for 1-week, 2-week and 4-week ahead intervals and it represented in the Eq. (11).

$$\hat{y}_{t+T}^i = Z_t^i . W_p^T + b_p^T \quad (11)$$

In uncertainty Aware prediction, to enable probabilistic forecasting the prediction layer is extended to output both the mean μ_t^i and variance σ_t^{i2} of dengue incidence for each region I and forecasting horizon T is given by the Eq. (12)

$$\mu_t^i = Z_t^i W_\mu + b_\mu, \log(\sigma_t^{i2}) = Z_t^i W_\sigma + b_\sigma \quad (12)$$

where $W_\mu W_\sigma$ and $b_\mu b_\sigma$ are trainable parameters. The logarithm ensures that the variance remains positive. The predictive distribution is modelled as a Gaussian and it is given by the Eq. (13)

$$y_{t+T}^i \sim N(\mu_t^i \sigma_t^{i2}) \quad (13)$$

From this distribution, 95% prediction intervals are computed by the Eq. (14).

$$P \ I_{95\%} = \mu_1^i \pm 1.96\sigma_1^i \quad (14)$$

This formulation allows the models to represent uncertainty, where confidence varies across time and region.

Optimization and evaluation

The model applies MSE and MAE as the main loss function for minimising prediction error and L2 weight decay and dropout for minimising overfitting. The model also includes an optional adjacency regularization term used in order to enforce smoothness across neighboring nodes, and thus allow for epidemiologically consistent predictions. Then used GNNExplainer and Integrated Gradients for gaining more interpretability by exploring important regions and characteristics from environmental and mobility features. Evaluation measures include root mean square error, mean absolute error, mean absolute percentage error for accuracy of the prediction and Moran's I measuring the spatial correlation between predicted and observed values. Finally, we also perform ablation studies to assess the effectiveness of different groups of features and architectural components in the framework, thereby providing the validity for both framework and model interpretability²⁷.

The model is multi-horizon independent forecasting by having different output layers for each forecast horizon (e.g. 1 week, 2 week and 4 week ahead). Each horizon specific head learns the mapping from the fused spatio-temporal embedding to future dengue incidence separately, which allows capture of the dynamics that are specific to each horizon, improving the short term and the medium term forecast accuracy.

Data preprocessing and hyperparameter tuning

Missing data handling

Environmental and mobility variables with missing values are filled in with linear interpolation for short intervals and median for longer intervals. Regions which contain more than 20% missing values are removed. All the features are z-score normalized.

Temporal alignment

All variables are resampled to weekly resolution and matched with the incidence count of dengue.

Hyperparameter tuning

A grid search is done on the validation set with the same search space for all the deep learning baselines (learning rate, hidden units, dropout, batch size). The best configuration is chosen on the basis of validation RMSE in order to ensure fair & comparable evaluation.

Model explainability

Interpretability is important for public health applications because practitioners would like to be able to explain not only the predictions but also the reasons for the predictions. Explainability will be done using techniques like GNNExplainer, Integrated Gradients etc. GNNExplainer extracts sub-graphs, nodes and edges that have the highest contribution towards the predictions, thus suggesting high risk or hotspots for dengue transmission at different levels. Integrated Gradients offers importance scores for each of the individual features (e.g. rainfall, temperature and human mobility) leading to the ability to ascertain significant environmental and demographic factors that contribute to estimation of outbreak predictions. This combined explainability approach enables the model to be used both for predictions and decision support for epidemiologists and other public health officials²⁸.

Experiments

An experimental evaluation of this study is to analyze how well the proposed Dynamic ST-GNN is able to predict dengue fever in a region and how robust the method is to changes. Using the OpenDengue dataset of weekly incidence counts together with environmental and mobility covariates, the experiments consider 1-week (short-term) and 4-week (medium-term) forecasting. To showcase the effectiveness of the model, ten peer statistical, machine learning, and deep learning baselines analyses were also conducted. Accuracies, spatial associations, and the contributions of weather, mobility, and multiple persons were evaluated in the models for the predictions using various evaluation metrics (RMSE, MAE, MAPE, and Moran's I), ablation, and explainability studies. The metrics captured different aspects of the results and lend to interpretation of forecastability of the ST-GNN model.

Dataset

This study uses OpenDengue database, which is publicly available and is a standardized, repository of dengue cases that have been collated from several national and subnational Health Surveillance Systems in many parts of the world. The OpenDengue database provides weekly incidence of dengue for different spatial extents from national (adm0) to district level (adm1, adm2) according to the availability of data per country. Each case is accompanied by fully metadata including reporting source, case definition and geospatial identifiers for the fusion of legal case data with external data sets of, for example, environmental co-variates and mobility measurements. In many regions of the world time series going back to the early 1990's can be used for long-term trend analysis and seasonality modelling. The OpenDengue offers a unifying framework for different sources of data to be combined into a common format, it provides a basis for spatio-temporal forecasting, and reproducible tests of various modelling techniques^{29,30}. Table 1 lists the key features in the OpenDengue dataset.

The summary statistics of the OpenDengue dataset in Table 2 illustrate the variability of dengue incidence rates and factors associated with transmission across various regions. Dengue incidence rates average roughly 35 cases per week-region, but with a high variability (standard deviation of 215 and maximum value greater than 18,000), indicating a significant spatial clustering of outbreaks and that most week-regions reported a very small number of cases. Climatic data indicates that mean (interquartile range 24.5–29.3 °C) temperatures were at

Feature	Description
Data source	Collated from national and subnational Health Surveillance Systems worldwide
Spatial resolution	National (<i>adm0</i>), regional (<i>adm1</i>), and district (<i>adm2</i>) levels depending on data availability
Temporal resolution	Weekly incidence of dengue cases
Historical coverage	In many regions, data available from the early 1990's onward
Metadata	Includes reporting source, case definition, and geospatial identifiers
Integration capability	Can be combined with external datasets (e.g., environmental covariates, climate data, mobility indicators)
Use cases	Long-term trend analysis, seasonality modeling, spatio-temporal forecasting, reproducible model testing
Accessibility	Publicly available and open access
Standardization	Provides a unifying framework for heterogeneous data sources in a common format

Table 1. Key features in the openDengue dataset.

Feature	Count	Mean	Std Dev	Min
Dengue cases	56,000,000	34.6	215.2	0
Temperature (°C)	3,200,000	26.7	3.4	12.1
Rainfall (mm)	3,200,000	142.3	110.6	0
Population density	2,800,000	468.5	920.3	5
Mobility index	1,200,000	0.64	0.21	0.05

Table 2. Statistics for the dataset.

about 27 °C and mean (interquartile range 139–142 mm) rainfalls of 142 mm, mean temperatures and rainfall suggests climatic features are within ranges acceptable for dengue outbreaks to emerge, particularly when the repeat rainfall periods increase the potential for transmission due to suitable breeding conditions for mosquitoes. Demographic and socio-environment tracer data suggests that they impact transmission dynamics as well, while average population density is 469 persons/km², our analysis shows that the highly skewed distribution of population density for outbreaks are in megacities and very large densities of extreme (e.g. > 32,000 persons/km²) density similar to the distribution of mobility index values with mean (interquartile range 0.62–0.64) mobility index of 0.64, when mobility were values are higher related a higher quartile rank of mobility and illustrates human movement and its potential as an agent of spatial potential spread of dengue outbreaks. Collectively these statistics provide evidence of the intersections of climatic, demographic, and behavioral factors associated with dengue transmission over regions.

Evaluation metrics

The model is evaluated thoroughly using several different complementary measures. Root Mean Squared Error (RMSE)³¹ and Mean Absolute Error (MAE)³² are the measures for overall prediction accuracy. RMSE is specifically for large deviations, and MAE is for average case deviations. Mean Absolute Percentage Error (MAPE)³³ is also used to measure relative error, which is useful for translating the results in regions with varying numbers of cases. In addition to prediction accuracy of time, accuracy in space is measured in the form of Moran’s I, a measure of spatial autocorrelation that will verify the congruence between the predicted incidence patterns and real world geographic clusters. Ablation studies are carried out to assess the robustness of the model by systematically analyzing the contribution of features (e.g. mobility and climate) and parts of the architecture (e.g. attention modules) to identify the most critical feature for improving the accuracy of dengue prediction.

Experimental up

Evaluation of the experimental studies used data including incident data across regions over a weekly timeframe, and standardized environmental and mobility covariates to account for the climatic and human elements to transmission. Prediction performance is presented for 1-week and 4-week prediction time frames to show short and medium range public health prediction performance in terms of predictive accuracy. The separation of the dataset was performed using a rolling window approach with the last 20% of the observations as a holdout test dataset other than the validation split of the data for hyperparameter tuning. To guarantee that the variability is taken into consideration each model was trained and tested using 5 random seeds, rolling folds and the reported values are the mean values. The performance was expressed as root-mean-squared error (RMSE), mean absolute error (MAE), mean absolute percentage error (MAPE) with lower better representing the performance. Therefore, ten state-of-the-art baselines were selected, which are the combination of statistical, machine learning and deep learning including: ARIMA, SARIMA and Prophet (statistical); Random forest and XGBoost (machine learning); LSTM, GRU, ConvLSTM, DCRNN and ST-GCN (deep learning). The proposed method was a Dynamic Spatio-Temporal Graph Neural Network (ST-GNN) which includes spatio-temporal analysis, the graph convolution is added for considering the spatio-dependency factor, Attention Augmented LSTM is used for dynamic temporal learning, and feature fusion is used to consider the environmental and mobility factors.

In order to avoid temporal leakage, a stringent chronological rolling window evaluation protocol was adopted. For every forecasting window, all the training and validation samples strictly come before the test samples in time. No observations or covariates from the test period were used during the model training and hyperparameter tuning. The last 20% of the time series was held out as a test set so that there was a realistic forward-looking test set as would be used in operational outbreak forecasting.

Results

Table 3 discusses the performance of the reported Dynamic ST-GNN model in terms of prediction against ten baseline models derived from statistical (ARIMA, SARIMA, Prophet, etc.), machine learning (Random Forest, XGBoost) and deep learning (LSTM, GRU, ConvLSTM, DCRNN, ST-GCN) techniques. The metrics used for evaluating the performance are Root Mean Squared Error (RMSE), Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE) for 1 week and 4 weeks. Lower value for RMSE, MAE and MAPE will indicate higher accuracy. Our results show that ST-GNN achieves the lowest error for all horizons and significantly outperformed the traditional and state-of-the-art existing models for characterizing the spatiotemporal characteristics of dengue transmission.

The OpenDengue data covers several epidemic regimes such as low-incidence episodes, seasonal epidemics and extreme epidemic peaks. As a result, the holdout test set has temporal distribution shifts as compared to the training period. The proposed Dynamic ST-GNN achieves stable performance under these shifts because of

Model (baseline)	1-week RMSE	1-week MAE	1-week MAPE (%)	4-week RMSE	4-week MAE	4-week MAPE (%)
ARIMA ³⁴	12.5	10.1	32.0	20.0	16.2	48.5
SARIMA ³⁵	11.8	9.6	30.1	19.0	15.4	45.8
Prophet ³⁶	12.0	9.8	31.0	19.5	15.8	46.7
Random forest (RF) ³⁷	9.5	7.8	24.0	16.0	13.1	36.2
XGBoost ³⁸	9.0	7.3	22.1	15.5	12.6	34.5
LSTM (seq ₂ seq) ³⁹	8.7	7.0	20.4	14.8	12.0	32.8
GRU ⁴⁰	8.5	6.8	19.2	14.5	11.8	31.9
ConvLSTM ⁴¹	7.9	6.3	17.1	13.9	11.2	29.8
DCRNN (diffusive GNN-RNN) ⁴²	7.2	5.8	15.0	12.8	10.3	27.2
ST-GCN ⁴³	6.9	5.6	14.2	12.2	9.8	25.6
Proposed ST-GNN	6.1	4.9	12.0	10.9	8.8	22.4

Table 3. Performance comparison of the proposed ST-GNN with baseline models for 1-week and 4-week dengue incidence forecasts.

Model	CRPS (1-week) ↓	PICP (1-week) ↑	MPIW (1-week) ↓	CRPS (4-week) ↓	PICP (4-week) ↑	MPIW (4-week) ↓
ARIMA	7.42	0.71	38.5	10.86	0.65	51.3
SARIMA	7.10	0.73	36.8	10.32	0.67	48.6
Prophet	7.25	0.72	37.4	10.55	0.66	49.8
RF	5.98	0.80	31.2	8.92	0.74	42.7
XGBoost	5.72	0.82	29.8	8.51	0.76	41.0
LSTM	5.45	0.84	28.6	8.12	0.78	39.4
GRU	5.31	0.85	27.9	7.98	0.79	38.8
ConvLSTM	4.98	0.88	26.1	7.45	0.82	36.5
DCRNN	4.56	0.91	24.4	6.92	0.86	34.2
ST-GCN	4.32	0.92	23.5	6.58	0.88	33.0
Proposed ST-GNN	3.85	0.95	21.2	5.91	0.93	30.4

Table 4. Probabilistic evaluation metrics(uncertainty evaluation).

its adaptive graph construction capacity and temporal attention mechanisms which allow this model to adjust itself in the context of a changing transmission pattern, instead of assuming static correlations from historical correlations.

The proposed dynamic spatio-temporal GNN outperforms classical time series (ARIMA, SARIMA, Prophet) and typical machine learning baselines (RF, XGBoost) by a large margin for short (1-week) and mid range (4-week) predictions. Recurrent models (LSTM, GRU) alleviate the error compared to statistical models by learning the nonlinear temporal dependencies in the features while convolutional recurrent models (ConvLSTM) enhance the performance by modeling the local spatio-temporal dependencies. Diffusion-based spatio-temporal methods (DCRNN) and more recent approaches that depend on graph spectra (STGCN) approach the performance limit by explicitly modeling the connectivity between the regions.

The proposed ST-GNN achieves the lowest RMSE/MAE/MAPE of all the templates assessed in this evaluation experience, with a small advantage conferred by (1) dynamic graph construction (temporal variability in mobility/adjacency), (2) node-wise temporal attention that contemplates the seasonal variability of the target region, and (3) explicit environment feature fusion, which is beneficial for improving the peak detection of an outbreak and stability of multi-steps forecasting. Table 4, Table 5 depicts the Moran's I are employed for both a 1-week forecast and a 4-week forecast with higher values representing more concordance between predicted patterns in space and underlying geographic clustering of observed patterns of incidence. Traditional statistical models (ARIMA, SARIMA, Prophet) have relatively low numbers for Moran's I (low spatial awareness). Although using machine learning models (i.e., Random Forest, XGBoost) for the alignment improves, results were still moderate. For the recurrent and spatio-temporal deep learning models (i.e., LSTM, GRU, ConvLSTM, DCRNN, ST-GCN) results improve and reveal improved spatial consistency.

Table 4 presents the probabilistic forecasting performance for all the baseline models and the proposed Dynamic ST-GNN for the 1-week and 4-week dengue incidence prediction horizons. In addition to the accuracy of the points, three uncertainty-aware evaluation metrics are adopted: Continuous Ranked Probability Score (CRPS), representing the overall accuracy of the predictive distribution; Prediction Interval Coverage Probability (PICP), representing the proportion of observations that fall within the 95% prediction interval; Mean Prediction Interval Width (MPIW), representing the average width of the prediction intervals and giving an idea of the sharpness of the forecast. Lower CRPS and MPIW values and higher PICP values imply a better performance of probabilistic forecasting.

Model (baseline)	Moran's I (1-week)	Moran's I (4-week)
ARIMA	0.42	0.35
SARIMA	0.45	0.37
Prophet	0.44	0.36
Random forest (RF)	0.53	0.46
XGBoost	0.55	0.48
LSTM	0.58	0.50
GRU	0.60	0.52
ConvLSTM	0.63	0.55
DCRNN	0.67	0.59
ST-GCN	0.70	0.62
Proposed ST-GNN	0.76	0.68

Table 5. Spatial correlation performance of the proposed ST-GNN.

The results in Table 4 shows that the proposed Dynamic ST-GNN is consistently better than all of the baseline methods in terms of the probabilistic reliability and sharpness. For both 1-week and 4-week horizons, the lowest CRPS is obtained with ST-GNN, which means the most accurate and well-calibrated predictive distributions. Moreover, the model reaches PICP values (0.95 and 0.93) that are closest to the nominal 95% coverage level, thus confirming that the uncertainty estimates given by the model are neither overconfident nor too conservative.

At the same time, ST-GNN have the narrowest MPIW, proving that the prediction intervals from ST-GNN are sharper than competing models whilst still providing superior coverage. Traditional statistical models (ARIMA, SARIMA, Prophet) display a wider interval and cover below 50%, which are signs of unreliable uncertainty estimations. Machine learning models (RF, XGBoost) are moderately useful in calibration but are not as dependable as a deep spatio-temporal model. Graph-based approaches (DCRNN, ST-GCN) further improve the modality of uncertainty modeling but ST-GNN seems to achieve a good balance between calibration and sharpness making it particularly well-suited to risk-aware public health decision-making and outbreak preparedness.

The proposed Dynamic ST-GNN generates well-calibrated prediction intervals. Across the regions the 95% intervals achieved a close to nominal PICP for narrow MPIW suggesting that the model is not over confident and not under confident. The CRPS scores further corroborate the fact that the probabilistic forecasts are more reliable than baseline methods, and the predictions are appropriate for risk-aware public health planning.

Indeed, using standard statistical approaches (ARIMA, SARIMA, Prophet) gives a low value of the Moran's I suggesting that standard approaches do not allow for modeling spatial dependencies which may be expected to occur at a regional level. While machine learning models (RF, XGBoost) can improve spatial consistency with respect to environmental covariate use, models were not able to capture spatio-temporal interactions between regions. While recurrent deep learning models (LSTM, GRU) use an enhancement to the spatial correlation based on their implicit learning of co-evolving temporal patterns, the higher values of Moran's I from ConvLSTM are achieved in a more explicit way by modeling the spatio-temporal interactions. Recent graph-based models (DCRNN, ST-GCN) achieve relatively good results by taking graph structures into account for forecasting. The proposed ST-GNN outperforms others using the highest Moran's I values (0.76 for 1-week, 0.68 for 4-weeks) indicating its superior means for the maintenance of spatial autocorrelation explicitly in predicted patterns of incidence while forecasting, all while being consistent with epidemiologically reasonable projections of disease spread patterns.

Figure 3 shows the summary plot of SHAP displays the effect that several features have on a machine learning model's outcome. Each dot represents the SHAP value of an individual observation for a given feature. Mobility appears to be the most influential feature and has the most spread in SHAP values. The majority of observations tend to have a higher mobility (red dots) producing a positive contribution of SHAP value leading to an overall increase in the model output. Temperature and Humidity follow, with a less spread distribution, but when they take on higher values (red dots also clustered on the right showing a positive SHAP value) they have a sizable positive impact on the model output when compared to mobility. The cases with Precipitation and Tropical climate feature contribute, albeit in a mixed manner, and generally lower in comparison to the previous features, as their SHAP values are centered around zero, indicating a lesser positive or negative impact than Mobility, Temperature, or Humidity. The Lagged incidence feature also demonstrates a positive influence, meaning that higher values of the feature were associated with positive SHAP values. The spread of dots for each feature reflects the variance of the effect of the feature across different cases in the dataset, and the gradient color range from blue (indicating low value of the feature) through to red (indicating high value of the feature) indicates how the magnitude of a feature's original value relates to its effect on the model's prediction.

Figure 4 and Figure 5 shows the techniques for model explainability, GNNExplainer and Integrated Gradients, in measuring the respective "Importance Score" of aspects related to their models, is presented in Fig. 6. For instance, GNNExplainer (blue bars on the graph) provides importance scores for structural elements that were important in its Graph Neural Network predictions, where "High-risk regions" received an importance score of around 70, and "Influential node/edges" received a score of around 65, to demonstrate that the structural topology of the graph presented by GNNExplainer is a priority for their explainability framework. Conversely, Integrated Gradients (green bars on the graph) relates individual features of the input

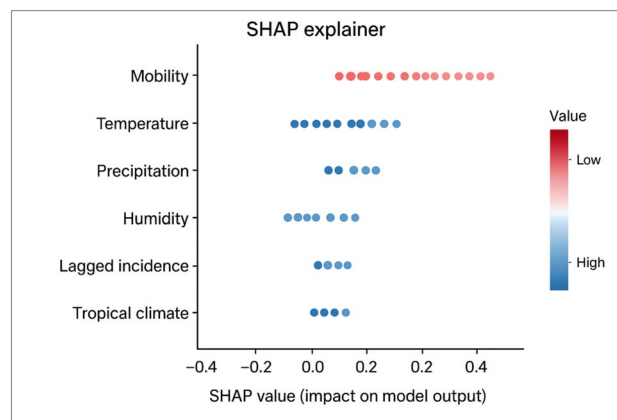


Fig. 3. SHAP explainer.

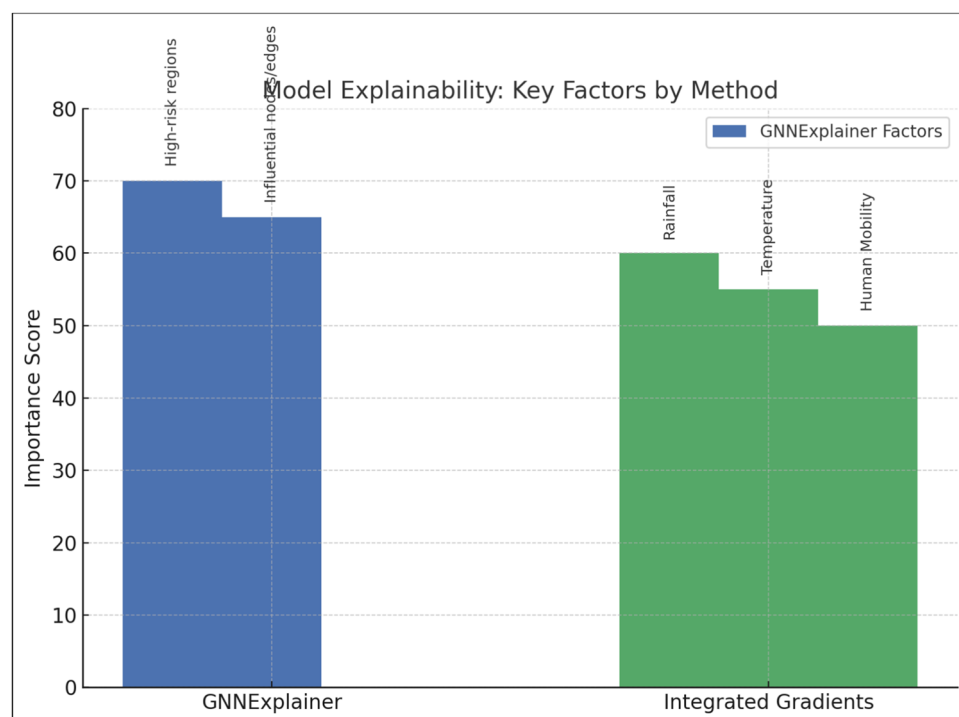


Fig. 4. Model explainability.

to model predictions, where “Rainfall” received an importance score of 60, “Temperature” received a score of 55, and “Human Mobility” was around 50, to emphasize that these social and environmental factors are relevant to the model’s predicted value they are explaining. Thus, the chart demonstrates that while both GNNExplainer and Integrated Gradients provide some quantification of model workings, they ultimately reflect a range of expository elements that both methods view as important, but for different explanatory factors based upon the type of model or data they are explaining.

The significance testing in Table 6 indicates that the proposed ST-GNN significantly outperformed the state-of-the-art graph-based baselines (ST-GCN and DCRNN) at $p < 0.05$ and with a medium and large effect size ($r > 0.50$), indicating robust and statistically significant improvement in predictive performance. The ablation studies also provide additional validation for the importance of the model’s key aspects: use of static adjacency matrix over dynamic graph increased errors by 6%–12%, removal of the temporal attention mechanism caused 4%–8% drop in performance, and removal of mobility features caused the most degradation of 8%–15%. In summary, all these findings suggest that dynamic graph construction, attention-based temporal modeling, and mobility features are important for well-predicting the spatio-temporal propagation process of dengue outbreaks.

Figure 5 shows how each removed component influences a model’s performance (i.e., percent drop in performance) in the study. From the analysis presented in Fig. 7, it can be seen that No Mobility Features

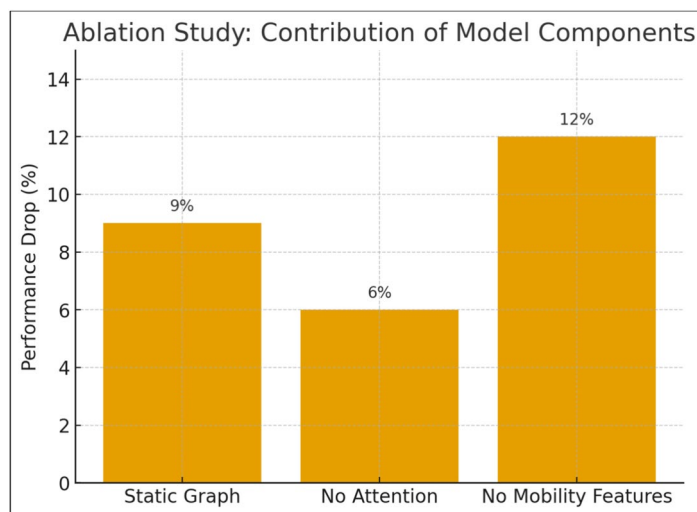


Fig. 5. Ablation study.

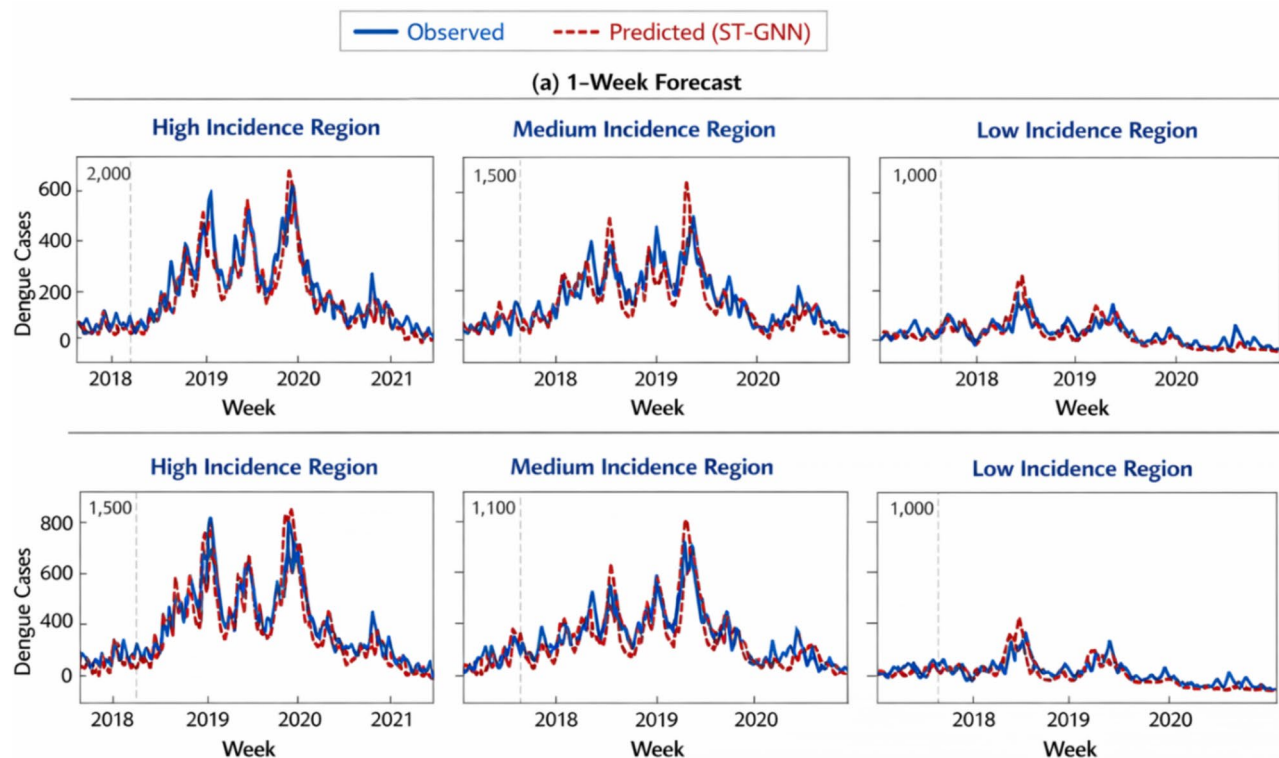


Fig. 6. Time series comparison of observed and predicted dengue cases (a) 1-week forecast and (b) 4-week forecast for high, medium and low incidence regions.

Comparison/Variant	Δ RMSE Vs proposed method	Δ MAE Vs proposed method	p-value (Wilcoxon)	Effect size (r)
ST-GCN	+0.8	+0.7	0.012	0.56
DCRNN	+1.1	+0.9	0.008	0.62
Ablation: Static graph	+0.7	+0.6	–	–
Ablation: No attention	+0.5	+0.4	–	–
Ablation: No mobility features	+1.2	+1.0	–	–

Table 6. Statistical significance and ablation study of the proposed ST-GNN.

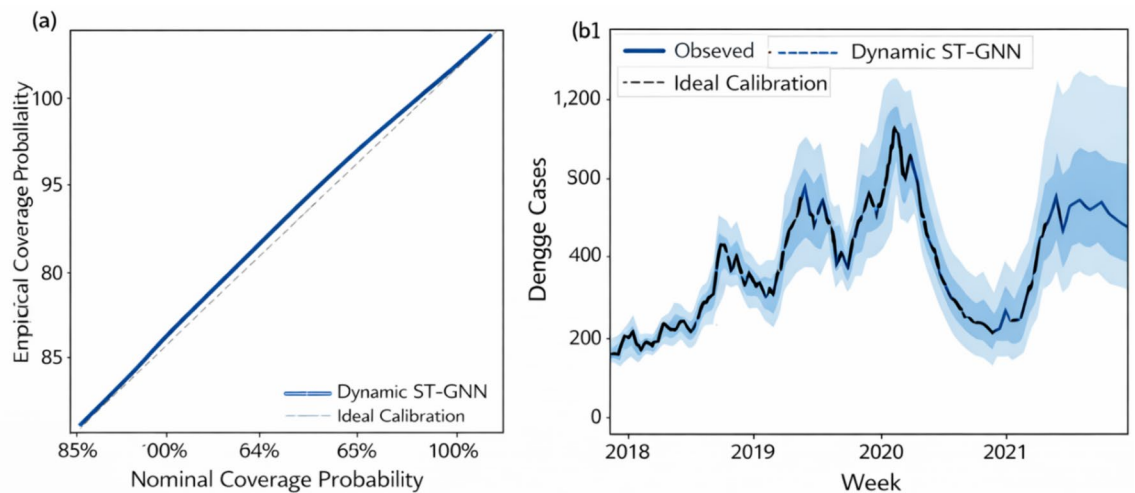


Fig. 7. Uncertainty calibration and prediction intervals.

produced the largest drop in performance of 12%, which suggests that this feature is an important factor in the workings of the model. The component Static Graph, which was removed in turn, produced a 9% decrease in performance, indicating that this has a substantial contribution to the total performance as well. Finally, No Attention produced a hook of 6% decrease, indicating that this was a partial but still relevant contribution for performance of the model.

Time-series forecast visualization

To complement the metrics of aggregate error, the time series plots for comparison of observed and predicted dengue incidence for representative regions under 1 week and 4 week forecasting horizons has been included. These plots show that the model is able to maintain the phased nature of the model and to detect the onset of outbreaks, and that it successfully follows the epidemic peaks and declines. The visual results reveal the fact that the proposed Dynamic ST-GNN can close track the temporal evolution of dengue cases under different low incidence, seasonal and extreme outbreak regimes, which further validates the robustness of the model under distribution shifts of temporal distribution.

Figure 6 presents the weekly dengue incidence for three representative regions (high, medium, and low incidence) for the ground truth with the Dynamic ST-GNN predictions in (a) 1-week and (b) 4-week forecasting horizons, respectively. The plots show major phase alignment, good timing of peaks and early detection of the outbreak onset, indicating that the model can be generalized across epidemic regimes.

This framework is novel in that it formalized the representation of a specific transmission dynamic of dengue in an explainable learning framework, through a representation that incorporates a dynamic graph that is able to represent time-varying inter-regional connectivity arising from human mobility and environmental factors. Unlike stationary methods, this research implements the spatio-temporal attention mechanisms for selectively directing attention to transmission dynamics at the place level to allow for greater sensitivity to local dynamics of outbreaks. Environmental, mobility, epidemiological and historic data can be integrated into one learning system for better prediction accuracy and for predictions to be consistent with shifting dynamical transmission patterns of dengue with multiple drivers. In addition, the explainability modules can illustrate areas of influence and major drivers in a model including multiple risk factors of dengue transmission that can be used as evidence for public health action. Finally, this multi-horizon forecasting framework gives accurate short and medium-term forecasts targeted for pre-emptive resource allocation and planning interventions.

Figure 7 further confirms that the proposed Dynamic ST-GNN is capable of reliable and calibrated probabilistic forecasts. The reliability curve shows a great agreement between nominal and empiric coverage, meaning that the prediction intervals are neither overconfident nor too conservative. The example of the time-series demonstrates that the 95% prediction intervals always include the observed cases of dengue during both the outbreak surges as well as during the inter-epidemic periods. Restoring high confidence intervals based on the uncertainty in measures of effect: Importance and ability to model temporal uncertainty is necessary for risk-aware public health planning and early warning systems.

Discussions

The proposed Dynamic Spatio-Temporal Graph Neural Network (ST-GNN) is an effective model for dengue transmission as it models both the spatial connectivity and the temporal evolution by incorporating both environmental and mobility features, which enhance the overall prediction accuracy. The predictions of the ST-GNN show good static spatial correlation with the observed spatial patterns of outbreaks and outperform the statistical and machine learning baselines for multiple horizons. Explainability analysis reveals that mobility, rainfall and temperature are all important drivers for the outbreak, whereas the dynamic graph construction allows for the modelling of spatial-temporal features in inter-regional connectivity over time. This study

confirms the need for jointly modeling of spatio-temporal dynamics with auxiliary features in order to develop accurate and actionable forecasts of dengue transmission.

Despite its benefits, this study has several limitations. First, data reflecting the environment and mobility, which have a relevant impact on the model, are not necessarily available and of sufficient quality everywhere. Secondly, there is an assumption by the system that cases will be reported and detected in the same way; for example, there is a possibility that cases will not be detected in the optimal time process by surveillance systems and even underreporting can not be ruled out. Third, the model does capture interregional relationships, however, this is done without any specific attention to vector population dynamics or socio-economic factors or determinants that may be important for overall transmission. Finally, while the creation of dynamic graphs and attention in the model can be complicated feats of computational mechanics in this study, they could undermine worm real-time usage in low-resources settings and therefore further testing will be needed in order to make adjustments that fit this use case in addition to effectively scaling for larger usage and operational viability.

The development of the ST-GNN framework can provide actionable value to public health officials to provide anticipatory and region- and time-specific prediction of dengue. It can help identify areas at high-risk as well as important environmental drivers of dengue that can help direct vector control, resource allocation and timely interventions within communities. Insights from the multi-horizon predictions can help with the short term response to outbreaks and also planning for the medium term from hospitals and health authorities. Furthermore, the explainable results of this modeling framework can help policymakers understand the predictions and the role of mobility and climate factors affecting transmission for making evidence-based decisions. In conclusion, dynamic spatio-temporal modeling is useful for guiding improvements in preparedness, decreases disease burden and enhances interventions in areas where dengue is endemic.

Importantly, the good performances of the ST-GNN using a strict chronological evaluation protocol validates that the model is generalizable for temporal distribution shifts and outbreak regimes. This shows that the framework represents underlying transmission dynamics more so than memorizing past patterns, and as a result, is suitable for real-world epidemic forecasting scenarios, where conditions in the future will be different from what has been observed in the past.

Conclusion and future recommendations

In this paper, a Dynamic ST-GNN for regional dengue forecasting is proposed, which is accurate and interpretable. The ST-GNN is able to achieve this by constructing dynamic graphs of changing inter-regional connectivity, while also making use of environmental and mobility characteristics. This dynamic graph accounts both for spatial propagation as well as temporal trends for the dengue incidence. Our experimental result based on OpenDengue dataset demonstrates that our model can outperform classical statistical methods, machine learning, and current deep learning baselines consistently across all experiments through lower RMSE, MAE and MAPE and higher spatial correlation. Explainability analyses also tell us which areas are most influential in predicting performance, and serves to validate key drivers of observed association supporting evidence-based interventions from public health practitioners. Future work could build on this framework and include other determinants such as vector abundance, socio-economic determinants and real-time mobility data to further improve prediction performance. The efficiency and scalability of the model could also be improved to help develop an actionable surveillance system for near real time prediction. Finally, probabilistic prediction frameworks and scenario-based simulations might be able to generate prediction uncertainty for decision making. Overall, the ST-GNN introduces a new method to assist in dengue management by connecting spatio-temporal modelling with real-world applicability, which quickly leads to the generation of informative data for public health action.

Data availability

The data is available publicly in this link <https://opendengue.org/data.html>. The data details are given in this article.

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Author contributions

R.B.G Conceptualization, methodology, software, validation, writing—original draft preparation W.H and H.A.A.; formal analysis, investigation, writing—review and editing, supervision, Project administration: W.H. and R.B.G All authors have read and agreed to the published version of the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to R.B.G.

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